AMC 8 · November Full Mock

Worked solutions • 25 questions • 40 minutes

Question 1 · Answer B

A stage crew paints five equal panels. Two panels are entirely blue, a third is half blue, and the other two contain no blue. What fraction of the whole wall is blue?

Choices: A 3/5 B 1/2 C 1/5 D 3/10 E 2/5

Count half-panels: five of the ten half-panels are blue, so the fraction is 5/10 = 1/2.

Answer: B · 1/2

Question 2 · Answer C

A rehearsal starts at 9:48 a.m., pauses for 7 minutes, and ends at 10:21 a.m. How many minutes were spent rehearsing?

Choices: A 33 B 40 C 26 D 19 E 27

The clock advances 12 minutes to 10:00 and 21 more to 10:21. Remove 7 from 33 to get 26.

Answer: C · 26

Question 3 · Answer E

A paper chain repeats red, red, gold, green, gold in that order. What is the position of the ninth gold link, counting the first link as position 1?

Choices: A 25 B 18 C 22 D 20 E 23

Four complete blocks have eight gold links and use 20 positions. The next gold is third in its block, at 23.

Answer: E · 23

Question 4 · Answer D

Four teams together collected 58 batteries. The first three collected 9, 14 and 16. How many more than the first team did the fourth collect?

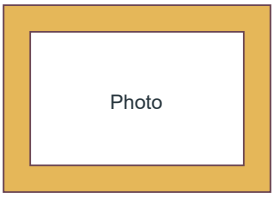
Choices: A 9 B 11 C 19 D 10 E 20

The fourth collected $58-(9+14+16)=19$. Compared with the first team that is $19-9=10$ more.

Answer: D · 10

Question 5 · Answer A

A photo measures 5 cm by 8 cm. A uniform 1 cm border surrounds its outside. What is the area of the border in square centimetres?



Choices: A 30 B 13 C 40 D 26 E 28

Adding a border on both sides makes a 7 by 10 rectangle. Subtract the photo: 70−40=30.

Answer: A · 30

Question 6 · Answer D

Three jars hold 12, 17 and 25 beads. A move takes one bead from a jar and puts it in another. What is the minimum number of moves to make all three quantities equal?

Choices: A 5 B 8 C 14 D 7 E 6

There are 54 beads, so each jar needs 18. The last jar has 7 too many. Moving six to the first and one to the second reaches the target in seven moves.

Answer: D · 7

Question 7 · Answer B

The tens and ones digits of a two-digit number add to 11. Reversing them increases the number by 27. What was the original number?

Choices: A 56 B 47 C 38 D 74 E 65

Reversal changes the value by nine times (ones digit − tens digit), so the ones digit is 3 larger. The digits add to 11, giving 4 and 7: the number is 47.

Answer: B · 47

Question 8 · Answer D

A 12 by 8 sheet is folded once so the two short edges coincide. A 2 by 3 rectangular notch is cut through both layers, away from the fold. What area remains after unfolding?

Choices: A 78 B 48 C 96 D 84 E 90

The fold creates two layers. Each loses 6 square units, so 96−12=84 remain.

Answer: D · 84

Question 9 · Answer B

A bag contains four plain counters and three striped ones. One counter is removed without looking, and then another is drawn. What is the probability that the first is plain and the second striped?

Choices: A 12/49 B 2/7 C 4/7 D 3/7 E 1/2

The first event has probability 4/7. After a plain counter leaves, three of the six left are striped. Multiply: (4/7)(3/6)=2/7.

Answer: B · 2/7

Question 10 · Answer E

A bicycle courier covers the first 3 km in 12 minutes and the next 5 km in 18 minutes. What was the average speed over the whole trip, in km/h?

Choices: A 20 B 17 C 15 D 12 E 16

Use total distance and total time, not the unweighted average of segment speeds. Eight kilometres in half an hour is 16 km/h.

Answer: E · 16

Question 11 · Answer A

On day 1 a club earns 3 points. Each following day it earns 2 more points than the previous day. On which day does the cumulative total first exceed 75?

Choices: A 8 B 10 C 6 D 7 E 9

The first n odd numbers starting at 3 sum to n(n+2). Seven days give 63; eight give 80, the first total above 75.

Answer: A · 8

Question 12 · Answer E

A menu offers three soups and four sandwiches. One sandwich is nut-free; exactly two soups are nut-free. How many soup-and-sandwich meals have at least one nut-free item?

Choices: A 5 B 8 C 6 D 10 E 9

There are 12 meals. Those with neither nut-free item use the one other soup and one of three other sandwiches: three meals. Thus 12−3=9.

Answer: E · 9

Question 13 · Answer A

Two identical cubes of side 3 cm are glued face to face. How many square centimetres of the resulting solid are exposed?

Choices: A 90 B 54 C 72 D 108 E 99

Two loose cubes have 12 faces. The two joined faces become internal, leaving 10 faces of area 9, for 90.

Answer: A · 90

Question 14 · Answer C

A drum taps at times 2, 8, 14, … seconds. A bell rings at 5, 14, 23, … seconds. At what time after 14 seconds do they next sound together?

Choices: A 23 B 29 C 32 D 36 E 41

Once the sounds coincide, their coincidence repeats after lcm(6,9)=18 seconds. Add 18 to 14 to get 32.

Answer: C · 32

Question 15 · Answer C

Two full identical water bottles and one empty bottle weigh 870 g. One full bottle and two empty bottles weigh 540 g. How many grams of water does a full bottle contain?

Choices: A 270 B 435 C 330 D 210 E 110

Both weighings contain three empty bottle shells. The first has exactly one additional bottleful of water, so subtract: 870−540=330 g.

Answer: C · 330

Question 16 · Answer A

Four boxes A, B, C, D hide one token. Their labels say, respectively: “in B or C”, “not in A”, “in D”, and “not in C”. Exactly three labels are true. How many possible hiding places are there?

Choices: A 2 B 3 C 0 D 4 E 1

If the token is in A, B, C or D, the numbers of true labels are respectively 1, 3, 2 and 3. B and D work, giving two possibilities.

Answer: A · 2

Question 17 · Answer D

A class scored an average of 14 on a quiz. When one score of 26 is removed, the remaining class average is 13. How many people were originally in the class?

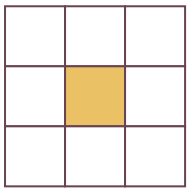
Choices: A 11 B 15 C 14 D 13 E 12

For n students, $14n-26=13(n-1)$. Expanding and simplifying gives $n=13$.

Answer: D · 13

Question 18 · Answer C

On a 3 by 3 board of unit cells, how many axis-parallel squares of any size contain the central cell?



Choices: A 9 B 8 C 6 D 4 E 5

There is one single-cell square, four different 2 by 2 squares, and one 3 by 3 square containing the centre. Their total is 6.

Answer: C · 6

Question 19 · Answer B

A jug is one-third full. After 8 cups are poured in, it is three-quarters full. How many cups would fill the jug from empty?

Choices: A 16 B 96/5 C 24 D 18 E 20

The added fraction is $3/4-1/3=5/12$. If 5/12 of capacity is 8 cups, the full capacity is $8\times12/5=96/5$ cups.

Answer: B · 96/5

Question 20 · Answer B

Among the positive integers less than 50, how many have exactly three positive divisors?

Choices: A 6 B 4 C 3 D 7 E 5

A number with three divisors must be a prime square: its divisors are $1, p, p^2$. The eligible primes are 2,3,5,7, giving four squares.

Answer: B · 4

Question 21 · Answer A

A solid 4 by 5 by 6 block made of unit cubes is painted on every outside face. How many unit cubes have paint on exactly two faces?

Choices: A 36 B 24 C 44 D 48 E 32

These cubes lie on edges, excluding corners. Four edges in each direction contribute $4[(4-2)+(5-2)+(6-2)]=36$.

Answer: A · 36

Question 22 · Answer D

Five bins stand in a row. Three indistinguishable red balls and two indistinguishable blue balls are placed one per bin. How many arrangements have the red balls in exactly two consecutive groups?

Choices: A 4 B 5 C 8 D 6 E 3

The ordered red group lengths are (1,2) or (2,1). The two blue balls must separate the groups; placing the extra blue before, between, or after gives three choices per order. Total 6.

Answer: D · 6

Question 23 · Answer A

A rooftop tank fills in 12 minutes with tap A and in 18 minutes with tap B. They run together for 4 minutes, then B shuts off. How many more minutes must A run?

Choices: A 16/3 B 8 C 6 D 4 E 20/3

Together the taps fill 5/36 per minute, hence 5/9 in four minutes. The remaining 4/9 takes $(4/9)\times12=16/3$ minutes with A.

Answer: A · 16/3

Question 24 · Answer A

A row starts as 14 white and 9 black counters. A move removes two counters: if their colours match, return a white counter; if they differ, return a black counter. After 22 moves how many black counters remain?

Choices: A 1 B 2 C 3 D 0 E 9

A move changes the black count by zero or two, so its parity stays odd. Each move reduces the total by one; 23 counters become one. It must be black.

Answer: A · 1

Question 25 · Answer C

Seven distinct whole numbers are chosen from 1 through 12. What is the smallest possible number of pairs of chosen numbers that differ by 1?

Choices: A 3 B 2 C 1 D 0 E 5

Five unchosen numbers can split chosen numbers into at most six runs. With seven chosen, at least one adjacent pair occurs. Choose 1,2,4,6,8,10,12 to attain exactly one.

Answer: C · 1